**POLYTECHNIQUE MONTRÉAL**
affiliée à l'Université de Montréal

# A lightweight coefficient extractor in C

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Ce mémoire intitulé :

**A lightweight coefficient extractor in C**

présenté par **William ZHANG**
en vue de l'obtention du diplôme de *Maîtrise ès sciences appliquées*
a été dûment accepté par le jury d'examen constitué de :

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## DEDICATION

*À tous mes amis du labos,*
*vous me manquerez...*

# ACKNOWLEDGEMENTS

Texte / Text.

# RÉSUMÉ

Le résumé est un bref exposé du sujet traité, des objectifs visés, des hypothèses émises, des méthodes expérimentales utilisées et de l'analyse des résultats obtenus. On y présente également les principales conclusions de la recherche ainsi que ses applications éventuelles. En général, un résumé ne dépasse pas trois pages.

Le résumé doit donner une idée exacte du contenu du mémoire ou de la thèse. Ce ne peut pas être une simple énumération des parties du manuscrit. Le but est de présenter de façon précise et concise la nature, l'envergure de la recherche, les sujets traités, les questions de recherche ou les hypothèses soulevées, les méthodes utilisées, les principaux résultats ainsi que les conclusions retenues. Un résumé ne doit jamais comporter de références ou de figures.

# ABSTRACT

Written in English, the abstract is a brief summary similar to the previous section (Résumé). However, this section is not a word for word translation of the abstract in French.

The abstract is a brief statement of the subject matter, objectives, research questions or hypotheses, experimental methods and analysis of results. It also presents the main research conclusions and their possible applications. In general, an abstract should not exceed three pages.

The abstract should provide an exact idea of the thesis or dissertation's contents and it cannot be a simple enumeration of the manuscript's parts. The goal is to precisely and concisely present the nature and scope of the research. An abstract should never include references or figures. If the thesis or the dissertation is in English, the résumé (French-language abstract) should come first followed by the abstract.

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# LIST OF SYMBOLS AND ACRONYMS

| | |
|---|---|
| IETF | Internet Engineering Task Force |
| OSI | Open Systems Interconnection |

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# CHAPTER 1 INTRODUCTION

## 1.1 Definitions and basic concepts

## 1.2 Problem statement

## 1.3 Research objectives

## 1.4 Thesis outline

# CHAPTER 2 LITERATURE REVIEW

## 2.1 Background

## 2.2 Related work

## 2.3 Summary

# CHAPTER 3 Setting the stage

**Functional convex programming** Users specify problems—by providing expression trees for each function—as the standard "functional" form for convex optimization

$$\begin{array}{ll} \text{minimize} & f_0(x;\theta) \\ \text{subject to} & f_i(x;\theta) \leq 0,\ i = 1,\ldots,m_1, \\ & g_i(x;\theta) = 0,\ i = 1,\ldots,m_2, \end{array} \tag{3.1}$$

with variable $x \in \mathbb{R}^n$ and parameter vector $\theta \in \mathbb{R}^p$ [1]. When $\theta$ is fixed, (3.1) is a problem instance, otherwise it defines a problem family.

**Quadratic conic programming** CVXPY transforms the problem in (3.1) into an equivalent quadratic conic program (QCP) of the form

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}x^T P x + q^T x \\ \text{subject to} & b - Ax \in \mathcal{K}, \end{array} \tag{3.2}$$

where $x \in \mathbf{R}^n$ is the variable, $\mathcal{K} \subseteq \mathbf{R}^m$ is a closed convex cone, and the *problem data* are $P \in \mathbf{S}^n_+$, $q \in \mathbf{R}^n$, $A \in \mathbf{R}^{m\times n}$, and $b \in \mathbf{R}^m$. CVXPY supports problems where the cone is a product of the zero cone ($\{0\}$), the nonnegative cone ($\mathbf{R}_+$), the second-order cone ($\mathcal{K}_{\text{soc}}$), the power cone ($\mathcal{K}_{\text{pow}}$), the exponential cone ($\mathcal{K}_{\text{exp}}$), and the cone of positive-semi-definite matrices ($\mathbf{S}_+$).

**Linear operator tree** CVXPY's canonicalization guarantees that all constraints and the objective function in (3.1) are affine functions of the variable $x$. An affine function can be expressed as the composition of linear functions and addition of constants. CVXPY represents each affine function as an *expression tree*, where each node is a linear function or *atom*.

**A first example** When you specify a single variable in CVXPY, such as

```
x = cp.Variable(4)
```

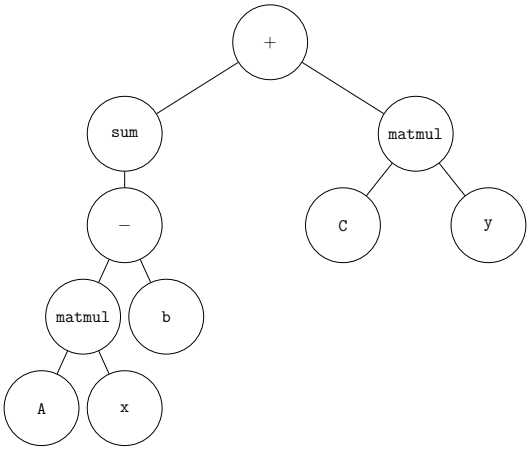



Figure 3.1 Expression tree for the affine expression `sum(A @ x - b) + C @ y`.

the internal representation is the identity linear operator of size $4 \times 4$:

$$\begin{array}{c} \begin{array}{cccc} x_1 & x_2 & x_3 & x_4 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

**A second example** When you specify a sum operator on a variable in CVXPY, such as

```
x = cp.Variable(4)
y = cp.sum(x)
```

the internal representation is a $1 \times 4$ matrix of all ones:

$$\begin{array}{c} \begin{array}{cccc} x_1 & x_2 & x_3 & x_4 \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

Applying this operator to the variable vector yields the sum:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_1 + x_2 + x_3 + x_4$$

**A matrix example** When you specify a matrix variable in CVXPY, such as

```
X = cp.Variable((3, 2))
```

the entries of the matrix are vectorized in column-major order, and the internal representation is the identity linear operator of size $6 \times 6$:

$$\begin{array}{c} \begin{array}{cccccc} X_{11} & X_{21} & X_{31} & X_{12} & X_{22} & X_{32} \end{array} \\ \left[\begin{array}{cccccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right] \end{array}$$

**A composition example** When you specify a more complex linear function in CVXPY, such as

```
x = cp.Variable(3)
A = np.random.randn(5, 3)
b = np.random.randn(5)
y = cp.sum(A @ x + b)
```

the internal representation is the composition of three linear operators:

- the matrix multiplication operator $A \in \mathbf{R}^{5\times 3}$,
- the vector addition operator $+b$ where $b \in \mathbf{R}^5$,
- the sum operator $\mathbf{1}^T \in \mathbf{R}^{1\times 5}$.

The constant $b$ is incorporated by augmenting the matrix with a column for the constant term. The resulting linear operator is $\mathbf{1}^T[A \mid b] \in \mathbf{R}^{1\times 4}$:

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \end{bmatrix}}_{\mathbf{1}^T} \underbrace{\left[\begin{array}{ccc|c} A_{11} & A_{12} & A_{13} & b_1 \\ A_{21} & A_{22} & A_{23} & b_2 \\ A_{31} & A_{32} & A_{33} & b_3 \\ A_{41} & A_{42} & A_{43} & b_4 \\ A_{51} & A_{52} & A_{53} & b_5 \end{array}\right]}_{[A|b]} = \begin{array}{cccc} x_1 & x_2 & x_3 & 1 \end{array} \left[\begin{array}{ccc|c} \sum_i A_{i1} & \sum_i A_{i2} & \sum_i A_{i3} & \sum_i b_i \end{array}\right]$$

**A multi-variable example** When you specify multiple variables and constraints in CVXPY, such as

```
x = cp.Variable()
y = cp.Variable(3)
z = cp.Variable(2)
constraints = [8*x == 28, cp.sum(y) + 5*cp.sum(z) == 58, z == 26]
```

the variables are concatenated into a single vector $[x, y_1, y_2, y_3, z_1, z_2]^T$. Each constraint is converted to the form $Ax + b = 0$, producing a linear operator with a constant column:

- `8*x == 28` becomes $8x - 28 = 0$:

$$\begin{array}{cccccc|c} x & y_1 & y_2 & y_3 & z_1 & z_2 & 1 \\ \hline 8 & 0 & 0 & 0 & 0 & 0 & -28 \end{array}$$

- `cp.sum(y) + 5*cp.sum(z) == 58` becomes $\sum y_i + 5\sum z_i - 58 = 0$:

$$\begin{array}{cccccc|c} x & y_1 & y_2 & y_3 & z_1 & z_2 & 1 \\ \hline 0 & 1 & 1 & 1 & 5 & 5 & -58 \end{array}$$

- `z == 26` becomes $z_1 - 26 = 0$ and $z_2 - 26 = 0$:

$$\begin{array}{cccccc|c} x & y_1 & y_2 & y_3 & z_1 & z_2 & 1 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 & -26 \\ 0 & 0 & 0 & 0 & 0 & 1 & -26 \end{array}$$

These are then stacked into a single linear operator:

$$\begin{array}{cccccc|c} x & y_1 & y_2 & y_3 & z_1 & z_2 & 1 \\ \hline 8 & 0 & 0 & 0 & 0 & 0 & -28 \\ 0 & 1 & 1 & 1 & 5 & 5 & -58 \\ 0 & 0 & 0 & 0 & 1 & 0 & -26 \\ 0 & 0 & 0 & 0 & 0 & 1 & -26 \end{array}$$

**The current approach: sparse matrix multiplication** CVXPY's current approach is to convert the linear operator tree into a single sparse matrix. The tree is traversed in

post-order: each leaf node returns its coefficient matrix, and each internal node composes its children's matrices via sparse matrix multiplication. For example, in the expression `sum(A @ x)`, the traversal first returns $A$ at the `matmul` node, then multiplies by $\mathbf{1}^T$ at the `sum` node, yielding $\mathbf{1}^T A$. The final matrix at the root is returned.

**Limitations of current approach** This approach requires storing intermediate sparse matrices at each node during traversal. For deep expression trees or large problem dimensions, this leads to significant memory overhead. Additionally, sparse matrix multiplication is computationally expensive and may produce matrices with more nonzeros than the operands, further increasing memory and runtime costs.

**Our approach: direct encoding of linear operators** We keep the coefficient matrix in CSR (Compressed Sparse Row) format and directly manipulate its index arrays without forming intermediate matrices. Each linear operator is implemented as a transformation on the CSR arrays `(indptr, indices, data)`. For example, the `sum` operator collapses all rows into row 0 by setting all entries of `indptr` to $[0, \texttt{nnz}]$, effectively mapping every row index to 0 without any matrix multiplication.

# CHAPTER 4 SECOND THÈME / SECOND THEME

Texte / Text.

# CHAPTER 5 TROISIÈME THÈMES

# CHAPTER 6 CONCLUSION

Texte / Text.

## 6.1 Synthèse des travaux / Summary of Works

Texte / Text.

## 6.2 Limitations de la solution proposée / Limitations

## 6.3 Améliorations futures / Future Research

Texte / Text.

## REFERENCES

[1] S. Boyd and L. Vandenberghe, *Convex optimization*. Cambridge university press, 2004.

# APPENDIX A DÉMO

Texte de l'annexe A. Remarquez que la phrase précédente se termine par une lettre majuscule suivie d'un point. On indique explicitement cette situation à LaTeX afin que ce dernier ajuste correctement l'espacement entre le point final de la phrase et le début de la phrase suivante.

# APPENDIX B ENCORE UNE ANNEXE / ANOTHER APPENDIX

Texte de l'annexe B en mode «landscape».

## APPENDIX C UNE DERNIÈRE ANNEXE / THE LAST APPENDIX

Texte de l'annexe C.